CHAPTER 14
The Repo Market

14.1 THE REPURCHASE MARKET

The word repo is a contraction of repurchase agreement which is a sale and simultaneous buy-back, for settlement at a later date, of the same security. In essence, a repo transaction is a temporary exchange of cash against a security which is referred to as collateral in this context. This can be compared to a mortgage in retail finance. The difference between the sale and the buy-back price of the security reflects an interest rate applied to the cash leg of the transaction, known as the repo rate. Because a security changes hands in the opposite direction of the cash, a repo transaction is structurally similar to a secured deposit. The repo market can be divided into three segments, the GC market, the specials market, and the interdealer market. The duration of a repo contract (i.e., the period between sale and buyback) can range from a single day to several months, but short duration trades dominate the overall market volume in both turnover and outstanding volume. Unlike the cash bond market, the settlement of the initial trade leg can be faster than the usual T+2 timing (of course, this does not apply in those bond markets that trade T+1 or even T+0 anyway, such as US Treasuries or UK Gilts). So-called open repo trades remain in place until cancelled, usually by the collateral lender. Such trades do not have an initial maturity and usually have variable interest rates.

The general collateral (or GC) market is where investors place or borrow cash against securities from broadly defined sets, such as ‘German, French or Dutch government bonds with a maturity between 1 and 10 years’. In such GC transactions, the party accepting the cash has discretion over what specific collateral is used in the transaction. Specification of collateral happens after the trade, including the repo rate, has been agreed between the parties. The focus of a GC transaction is the cash leg and GC repo is used both by cash lenders and borrowers to transact in a secure way. The GC repo rate is generally lower than the unsecured rate that would apply between the two counterparties which reflects the lower credit risk. As a matter of fact, the volume of unsecured lending between banks has declined drastically in recent years, with a corresponding increase in secured (repo) interbank lending. This shift in transaction terms results partly from increased risk awareness at banks, but also from the regulatory treatment of interbank risk on bank balance sheets.

In the specials market the security used as collateral is specified at the time the transaction is agreed. This market segment is also known as the specific market. A security is called special in the repo market when cash lenders are willing to accept a repo rate below, sometimes substantially below, the GC rate in exchange for that security. Typically this will be the case when there is a large demand to borrow that security, for instance for delivery to a futures exchange, or for market-making purposes.

Lastly, there is an interdealer market that forms the bridge between the GC and specials markets. A given security may have been lent out as GC for a longer term but since then become more sought after. The purpose of the interdealer market is to find such positions and use them to relieve shortages in the security. One reward for this service is the spread between the GC and specials rates.

There are two reasons why the repo market is so important. The first is that it is now much larger than the unsecured bank lending market. Repo reduces the credit risk in lending because collateral is available in case the borrower of the cash defaults. Second, repo makes it possible to sell securities one does not own. In the fixed income world, dealers generally are the counterparty to their customers, so a buy request from a customer is filled by selling the security to the customer with the dealer as the seller. This is very different from the equity world where the buy request is usually just forwarded to the next exchange. The dealer can only deliver the security the buyer asks for if the dealer either owns is or has a way to borrow it. Given the hundreds of bond issues traded actively in most bond markets, holding each of them in inventory would be prohibitively expensive. With an active repo market, the dealer can simply borrow a bond requested by a customer and buy the bond back later in the market or from another customer when it becomes available. In this case, the driving force of the trade is not a cash-rich institution wishing to place cash as in a deposit. Instead, it is a dealer wishing to borrow a security. Hence, traders speak of ‘buying a security in repo’ instead of calling the same transaction ‘lending cash in repo’.

There are different GC rates for different collateral standards. For instance, Italian GC (where any Italian government bond is elegible collateral) trades at a marginally higher rate than German GC because Italian government bonds are seen as having slightly higher credit risk.

However, when there is high demand for borrowing a particular security, the security is said to be special and a lower rate is being paid on the cash that is being lent than in GC transactions. This is because the trader lending the cash is willing to accept a lower income on the cash in return for having that particular security. A given security is said to be ‘20 bp special’ when the cash rate at which it can be bought (borrowed) is 20 basis points lower than the GC rate. A security is usually special only for certain time horizons, e.g., until the next futures contract delivery or until a new bond is being auctioned. Hence, an issue can be ‘100 bp special in overnight’ and ‘10 bp special in 1 month’, meaning that the repo rate for borrowing the issue for one day is 100 basis points below the overnight GC rate but when the issue is borrowed for a whole month, the rate is only 10 bp below the one month GC rate. Because repo is generally quoted in terms of collateral borrowing, the bid rate in repo is higher than the offer rate.

Repo rates depend on the type of clearing used. Centrally cleared repos imply lower balance sheet cost, and therefore structurally lower repo rates, than bilateral repos. A specific form of repo where the securities do not change hands but are instead held at a trusted third party (tri-party repo) has a similar effect as central clearing and is becoming more dominant.

14.2 HAIRCUT

Although the repo rate is the main economic variable in repo transactions, another important factor is the so-called haircut. The haircut is a percentage amount of collateral notional that is disregarded for the purpose of calculating the amount of cash that can be borrowed against this collateral. For instance, at a haircut of 1% and with collateral worth JPY 10bn (market value), only 9.9bn in cash can be borrowed. The haircut protects the cash lender against variations in the collateral value in the period it takes to either unwind the repo transaction, or replenish collateral. The amount of haircut, therefore, depends on collateral quality and the price volatility of the collateral. Government bond collateral usually has haircuts of 0% to 1% depending on maturity while equity haircuts can be around the 50% level.

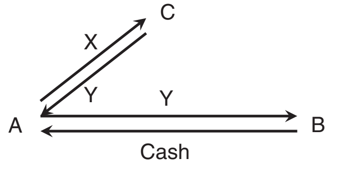
Haircuts can also depend on the relative credit risk of the contracting parties. Because a higher haircut means that the securities lender transfers more value to the cash lender than the cash amount received, the securities lender is exposed to the risk that the cash lender defaults and the securities lender faces losses on the extra securities lent. This setup is the reason why central banks (who, in theory, cannot default on commitments in their own currency) can impose high haircuts on their counterparties. This ability to enforce asymmetric terms on repo counterparties is important because central banks tend to face wrong-way risk in the securities they receive. This concept refers to the high correlation between the value of the collateral, and the stability of the borrowing bank. Because banks are exposed, in various ways, to the borrowers that issue the collateral they use for funding, the value of the collateral is likely to decline at the same time as the borrowing bank becomes less sound, i.e., when the collateral is most important.

Usually, repo counterparties agree to revalue the collateral on a daily basis and to post or return collateral depending on the observed price changes. This is one example of mark-to-market.

14.3 VARIATIONS OF REPURCHASE TRANSACTIONS

Aside from straight repo, there are a number of other ways of achieving the same effect. The problem with repo is that in many markets the legal system does not recognise that lending the securities and returning them is the same transaction. This means that one counterparty in the trade is not completely covered in case the other side defaults. This problem is particularly important in cross-border lending. There are other transactions that are economically equivalent to repo, but have different legal structures. These are:

Securities lending Almost like repo without the cash leg, the lender lends the security in return for a fee. The fee is calculated from an interest rate that is effectively the spread between GC and the special rate of the security. Securities lending can be collateralised so that the securities lender receives another security for the duration of the lending operation. Such a transaction is also known as a collateral swap, which can have the purpose of collateral transformation. For instance, the holder A of a security X may wish to borrow cash from a lender B with particular lending standards that do not permit X as collateral. This might not be because X is of low quality but perhaps because X is not denominated in a particular currency, or located in a particular CSD.



The borrower A can use a collateral swap with a third party C to transform the collateral X into collateral Y which is acceptable to B. In this case, the funding cost for A is equal to the sum of the repo rate charged by B and the lending fee charged by C. For C, normally a large asset manager, the lending fee in effect increases the return on asset Y.

An important case of collateral upgrades was the Term Securities Lending Facility (TSLF) announced by the Fed in June 2009. Under the TSLF, primary dealers could borrow US Treasury securities from the Fed's SOMA portfolio against posting investment grade corporate and mortgage bonds, for a fee that was set at auction. In the middle of the financial crisis, this programme served to increase the quality of the collateral available for private sector repos, and so helped to improve the redistribution of cash between private sector entities.

Sell-and-buyback The repo transaction is documented as an outright sale and a subsequent buyback at a later time. The price at which the buyback is executed is fixed at the time of the initial sale and reflects the repo rate at which the bond is trading. The cash flows and securities flows are the same as in a normal repo, but some problems are introduced. For instance, the seller must realise any capital gains or losses on the securities and faces the possibility that in the case of default, the securities sold will not be returned. Instead, other securities or cash may be returned.

Tri-party repo In this transaction, the collateral securities are not transferred to the cash lender and are instead held in trust by a third party. Usually, that third party is a depository institution or a clearing house. This tri-party repo can eliminate the need to actually transfer securities if both the cash lender and cash borrower are able to use this third party as their clearer. Although tri-party is not really useful for the purpose of borrowing securities, it is much cheaper than standard repo and so more efficient for secured cash lending.

Straight repo is used by many central banks to create money. The ECB main refinancing and longer term lending operations are generally repos although the ECB can also use outright buying and selling of securities to control liquidity. The Bank of

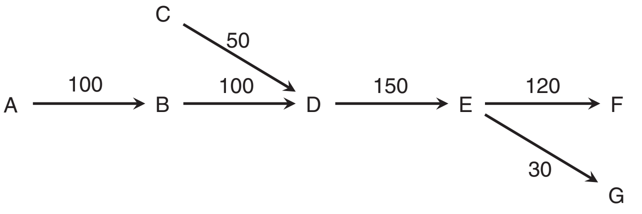


FIGURE 14.1 Example of rehypothecation of a single security. Letters designate counterparties and numbers the amount of collateral posted.

Japan uses repos (gensaki) together with outright buying and selling (formerly known as rinban). The Fed, on the other hand, mainly uses outright purchases and sales. When a central bank uses repo as the main liquidity tool, banks looking to fund in that currency usually have to have a branch in the respective country in order to be able to participate.

14.4 REHYPOTHECATION

It is possible that the counterparty receiving collateral uses the same collateral for another repo transaction until it is due to be returned. In some cases, this can lead to long chains of transactions where the same collateral is reused several times, as shown in Figure 14.1. This is known as rehypothecation and sometimes viewed as problematic. Note that collateral received can be fully or partly rehypothecated, and rehypothecated collateral can come from multiple sources. As long as all transactions settle and mature as planned, rehypothecation presents no issue. A problem arises when one of the counterparties in the chain of rehypothecation trades fails.

Although a counterparty failure presents negligible credit risk due to the collateralisation of a repo trade, it can cause delays in the return of collateral. In the example of Figure 14.1, failure of F could delay the return of collateral to E which would then propagate a failure all the way back to A and C, the original suppliers of the collateral. Note that A and C need not be aware of the existence of F, much less its economic situation. In case of a failure of G, an additional question becomes obvious: the collateral held could have come completely from A, or completely from C, or be a mixture of both. The extent to which A and C should be affected is not clear. A failure of F would clearly have to affect both.

Regulators are now looking into ways to address these issues, and in particular limit the number of links in such chains. The example shows why this is not a trivial issue. E receives collateral from A and C, and in doing so is the third link in the chain starting at A but only the second in the chain starting at C. While modern securities clearing systems can easily track beneficial ownership of individual pieces of securities, tracking such rehypothecation chains would require much more data processing.